JOIN Statements

Objectives

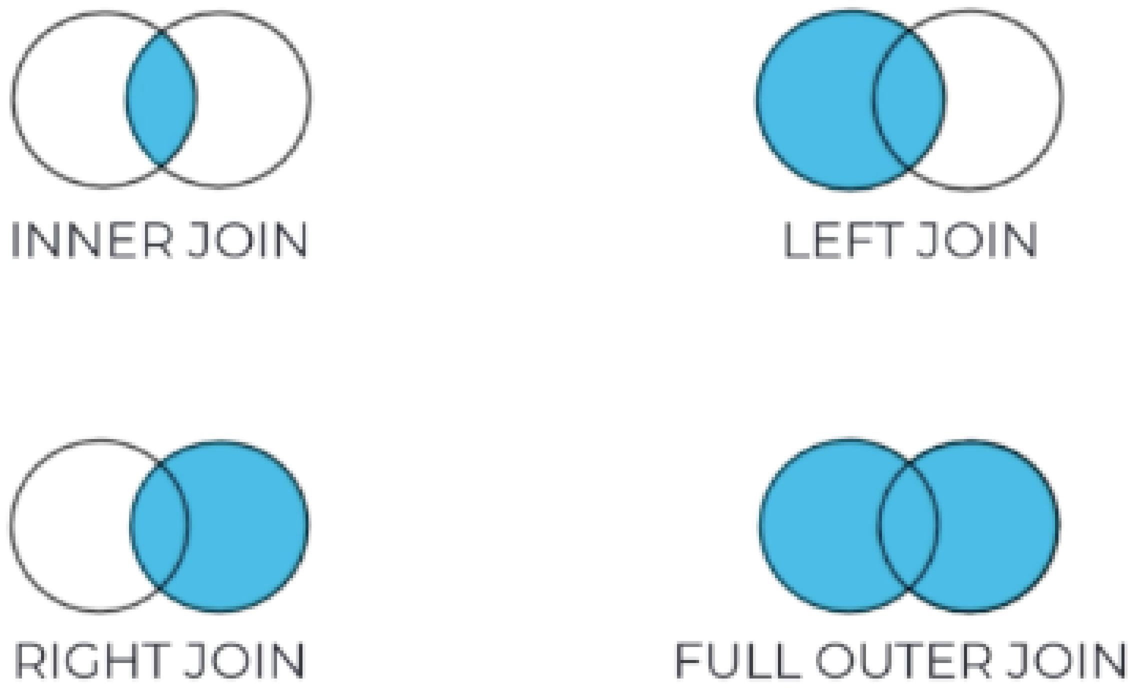
- Understand the syntax and structure of different types of **JOIN** statements in MySQL.
- Use **INNER JOIN** to retrieve matching records from two or more tables.
- Implement **LEFT JOIN** and **RIGHT JOIN** to retrieve unmatched records from one side of the join.
- Simulate **FULL OUTER JOIN** using **UNION** in MySQL.
- Discuss the role of **primary and foreign keys** to establish relationships between tables during joins.

What is JOIN?

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.
- It is a means of combining data in fields from two tables by using values common to each table.

Types of JOIN

1. Inner Join
2. Left Join
3. Right Join
4. Full Outer Join



What is MySQL Inner Join?

- The INNER JOIN keyword selects records that have matching values in both tables.

employee

emp_id	full_name	hire_date
1	Richard	2015-04-17
2	Hugo	2017-07-15
3	Melissa	2017-07-15
4	Heidi	2019-10-12
5	Polly	2015-04-17
6	James	2019-10-12
7	Lisa	2019-10-12
8	Michael	2015-04-17
9	Harry	2017-07-15
10	Julie	2019-10-12

salary

emp_id	prior_salary	current_salary
2	30000	32000
4	35000	35000
6	25000	25000
8	40000	45000
10	50000	50000
12	47000	50000
14	32000	35000
16	39000	39000
18	51000	52000
20	41000	41000

emp_id	full_name	hire_date	prior_salary	current_salary
2	Hugo	2017-07-15	30000	32000
4	Heidi	2019-10-12	35000	35000
6	James	2019-10-12	25000	25000
8	Michael	2015-04-17	40000	45000
10	Julie	2019-10-12	50000	50000

Sample Table

Orders Table

OrderID	CustomerID	EmployeeID	OrderDate	ShipperID
10308	2	7	1996-09-18	3
10309	37	3	1996-09-19	1
10310	77	8	1996-09-20	2

Customers Table

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

MySQL Inner Join Example

- ```
  SELECT Orders.OrderID, Customers.CustomerName
  FROM Orders
  INNER JOIN Customers ON Orders.CustomerID =
  Customers.CustomerID;
  ```

- **Note:** The INNER JOIN keyword selects all rows from both tables as long as there is a match between the columns. If there are records in the "Orders" table that do not have matches in "Customers", these orders will not be shown.
  ```

MySQL Inner Join Example

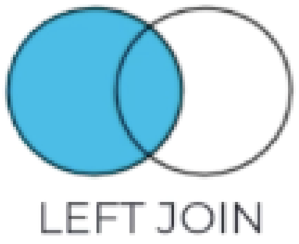
- ```
  SELECT Orders.OrderID, Customers.CustomerName,
  Shippers.ShipperName
  FROM ((Orders
  INNER JOIN Customers ON Orders.CustomerID =
  Customers.CustomerID)
  INNER JOIN Shippers ON Orders.ShipperID =
  Shippers.ShipperID);
  ```
  ```

MySQL Left Join

- The LEFT JOIN keyword returns all records from the left table (table1), and the matching records (if any) from the right table (table2).

employee

emp_id	full_name	hire_date
1	Richard	2015-04-17
2	Hugo	2017-07-15
3	Melissa	2017-07-15
4	Heidi	2019-10-12
5	Polly	2015-04-17
6	James	2019-10-12
7	Lisa	2019-10-12
8	Michael	2015-04-17
9	Harry	2017-07-15
10	Julie	2019-10-12



salary

emp_id	prior_salary	current_salary
2	30000	32000
4	35000	35000
6	25000	25000
8	40000	45000
10	50000	50000
12	47000	50000
14	32000	35000
16	39000	39000
18	51000	52000
20	41000	41000

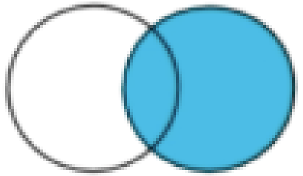
emp_id	full_name	hire_date	prior_salary	current_salary
1	Richard	2015-04-17	*null*	*null*
2	Hugo	2017-07-15	30000	32000
3	Melissa	2017-07-15	*null*	*null*
4	Heidi	2019-10-12	35000	35000
5	Polly	2015-04-17	*null*	*null*
6	James	2019-10-12	25000	25000
7	Lisa	2019-10-12	*null*	*null*
8	Michael	2015-04-17	40000	45000
9	Harry	2017-07-15	*null*	*null*
10	Julie	2019-10-12	50000	50000

MySQL Right Join

- The RIGHT JOIN keyword returns all records from the right table (table2), and the matching records (if any) from the left table (table1).

employee

emp_id	full_name	hire_date
1	Richard	2015-04-17
2	Hugo	2017-07-15
3	Melissa	2017-07-15
4	Heidi	2019-10-12
5	Polly	2015-04-17
6	James	2019-10-12
7	Lisa	2019-10-12
8	Michael	2015-04-17
9	Harry	2017-07-15
10	Julie	2019-10-12



RIGHT JOIN

salary

emp_id	prior_salary	current_salary
2	30000	32000
4	35000	35000
6	25000	25000
8	40000	45000
10	50000	50000
12	47000	50000
14	32000	35000
16	39000	39000
18	51000	52000
20	41000	41000

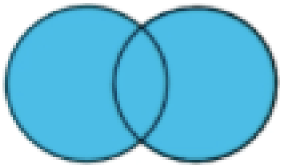
emp_id	prior_salary	current_salary	full_name	hire_date
2	30000	32000	Hugo	2017-07-15
4	35000	35000	Heidi	2019-10-12
6	25000	25000	James	2019-10-12
8	40000	45000	Michael	2015-04-17
10	50000	50000	Julie	2019-10-12
12	47000	50000	*null*	*null*
14	32000	35000	*null*	*null*
16	39000	39000	*null*	*null*
18	51000	52000	*null*	*null*
20	41000	41000	*null*	*null*

Full Outer Join

- Full JOINs combine both left and right joins by returning all rows from both tables, as long as there is at least one match between them.

employee

emp_id	full_name	hire_date
1	Richard	2015-04-17
2	Hugo	2017-07-15
3	Melissa	2017-07-15
4	Heidi	2019-10-12
5	Polly	2015-04-17
6	James	2019-10-12
7	Lisa	2019-10-12
8	Michael	2015-04-17
9	Harry	2017-07-15
10	Julie	2019-10-12



FULL OUTER JOIN

salary

emp_id	prior_salary	current_salary
2	30000	32000
4	35000	35000
6	25000	25000
8	40000	45000
10	50000	50000
12	47000	50000
14	32000	35000
16	39000	39000
18	51000	52000
20	41000	41000

emp_id	full_name	hire_date	prior_salary	current_salary
1	Richard	2015-04-17	*null*	*null*
2	Hugo	2017-07-15	30000	32000
3	Melissa	2017-07-15	*null*	*null*
4	Heidi	2019-10-12	35000	35000
5	Polly	2015-04-17	*null*	*null*
6	James	2019-10-12	25000	25000
7	Lisa	2019-10-12	*null*	*null*
8	Michael	2015-04-17	40000	45000
9	Harry	2017-07-15	*null*	*null*
10	Julie	2019-10-12	50000	50000
12	*null*	*null*	47000	50000
14	*null*	*null*	32000	35000
16	*null*	*null*	39000	39000
18	*null*	*null*	51000	52000
20	*null*	*null*	41000	41000

Full Outer Join Syntax

- *SELECT * FROM table1 FULL OUTER JOIN table2 ON table1.id = table2.user_id*

Outer Join vs. Cross Join

- A CROSS JOIN produces a cartesian product between the two tables, returning all possible combinations of all rows.
- It has no ON clause because you're just joining everything to everything.
- A FULL OUTER JOIN is a combination of a LEFT OUTER and RIGHT OUTER JOIN.
- It returns all rows in both tables that match the query's WHERE clause, and in cases where the ON condition cannot be satisfied for those rows it puts NULL values for the unpopulated fields.

Key Notes

1. The four most common join types are:

- Inner Join
- Left Join
- Right Join
- Full Outer Join

2. To join data, you need to establish the fields or fields used to link the tables together.

3. The syntax for executing each type of join is similar.

The Role of Primary Keys and Foreign Keys

Primary Keys

- A primary key is the column or columns containing values that ***uniquely identify*** each row in a table.

Primary Key

Employee ID	Employee Name	Job Title	Contact No
1	Sally	Finance Analyst	0123456789
2	John	IT Administrator	0987543210
3	Krupa	Sales Executive	0987656789

ID	Year	Salary	Employee ID
1	2020	55000	2
2	2020	60000	3
3	2020	65000	1
4	2021	68000	1
5	2021	70000	2
6	2021	65000	3

Primary Keys

Employee ID = Primary Key

Employee ID	Employee Name	Job Title	Contact No
1	Sally	Finance Analyst	0123456789
2	John	IT Administrator	0987543210
2	Krupa	Sales Executive	0987656789

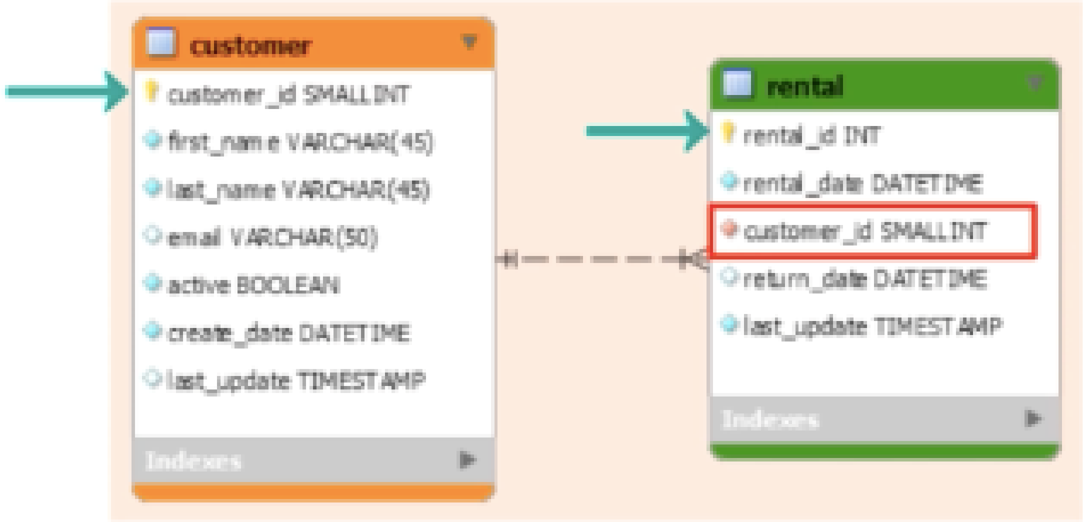
Rules in Creating Primary Key

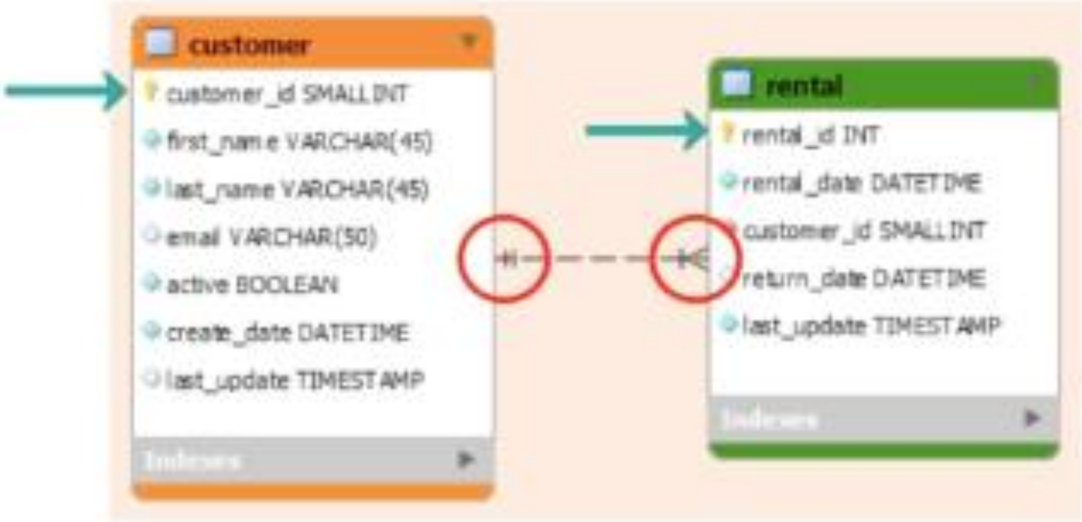
1. Must contain unique values
2. Cannot contain NULL values
3. A table can only have 1 primary key

Foreign Keys

- A *foreign key* is a column or columns in a database that (e.g. table_1.column_a) that are linked to a column in a different table (table_2.column_b).
- The existence of a foreign key column establishes a *foreign key constraint* – a database rule that ensures that a value can be added or updated in column_a only if the same value already exists in column_b.

Primary and Foreign Keys





Primary and Foreign Keys

A table can have more than 1 foreign key where each foreign key references to a primary key of different parent tables.

Key Notes

- A primary key is a column or multiple columns in a table that enables each row to be uniquely identified
- A foreign key is a column or columns in a table that provides a link between data in two tables
- A primary key can be joined to the foreign key in a join condition
- A primary key must be unique and cannot contain NULL values
- A foreign key does not need to be unique and can contain NULL values.

Summary of Key Roles

- **Primary key** ensures that each record in a table is unique, enabling accurate and consistent joins.
- **Foreign key** links tables together, facilitating relational database queries and enforcing referential integrity.
- During **joins**, the foreign key in one table is usually matched with the primary key in the other table to combine related rows across multiple tables.
- Without proper primary and foreign keys, joins would be less effective, and data integrity could be compromised.

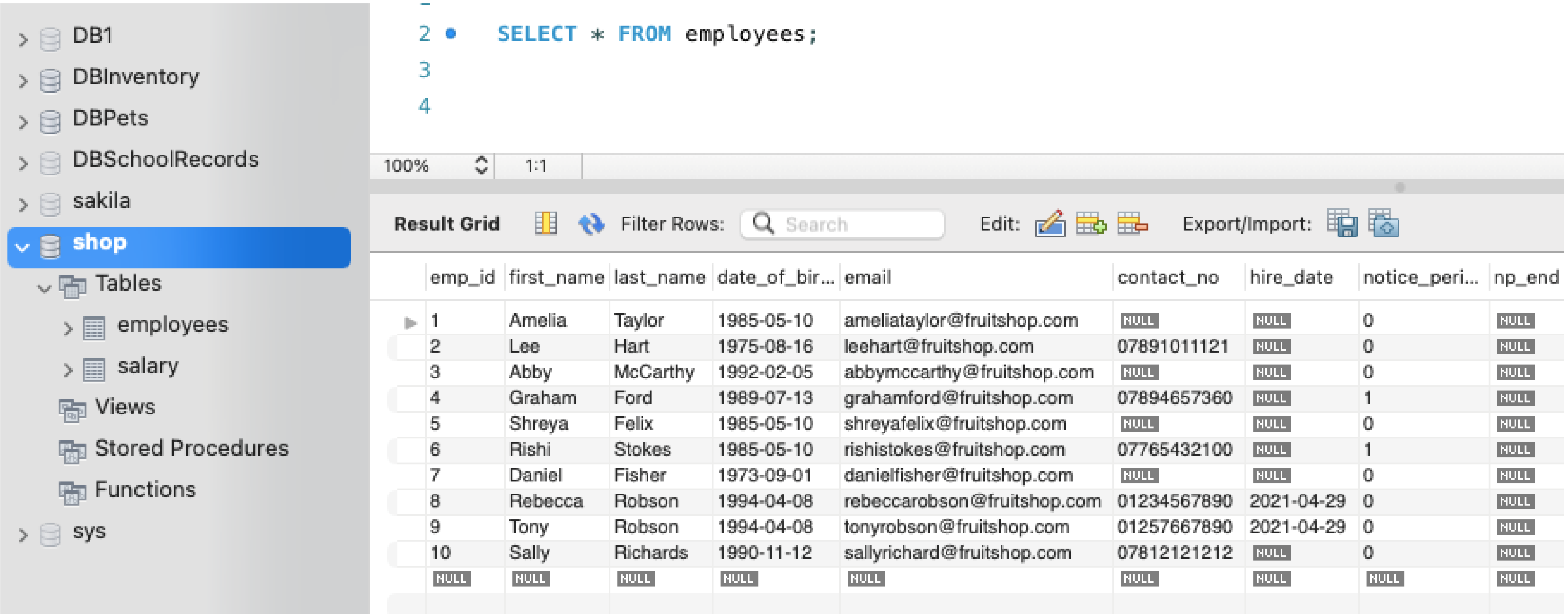
What is data integrity in database?

- **Data integrity** in a database refers to the accuracy, consistency, and reliability of the data stored.
- It ensures that the data remains correct and unchanged during operations such as inserts, updates, and deletions, and across multiple transactions or database operations.

Joining Tables

Overview of the Database

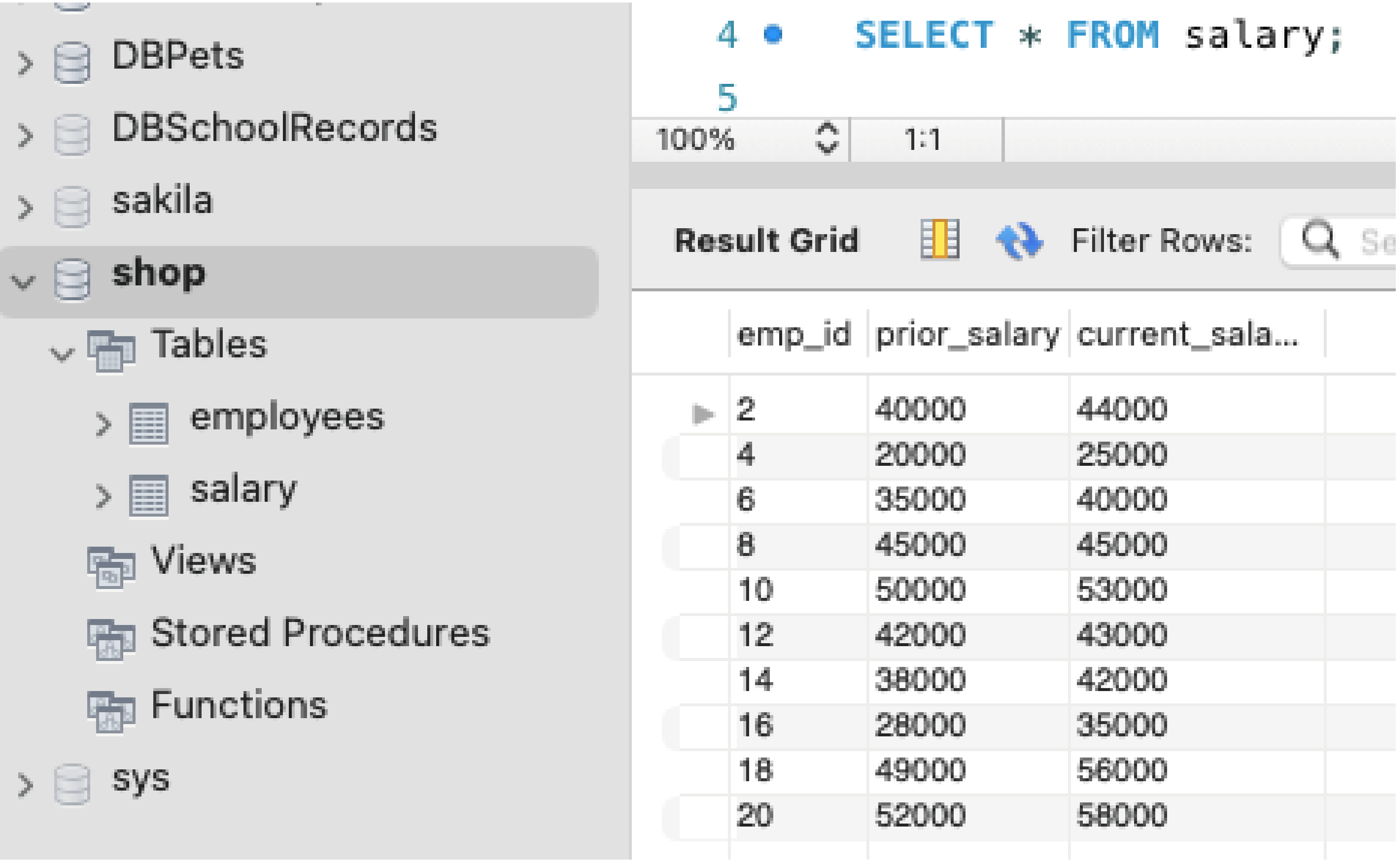
- Database Name: Shop
- Tables: employees, salary



emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_end
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Overview of the Database

- Database Name: Shop
- Tables: employees, salary



Overview of the Database

```
2 •   SELECT * FROM employees;
```

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_en
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

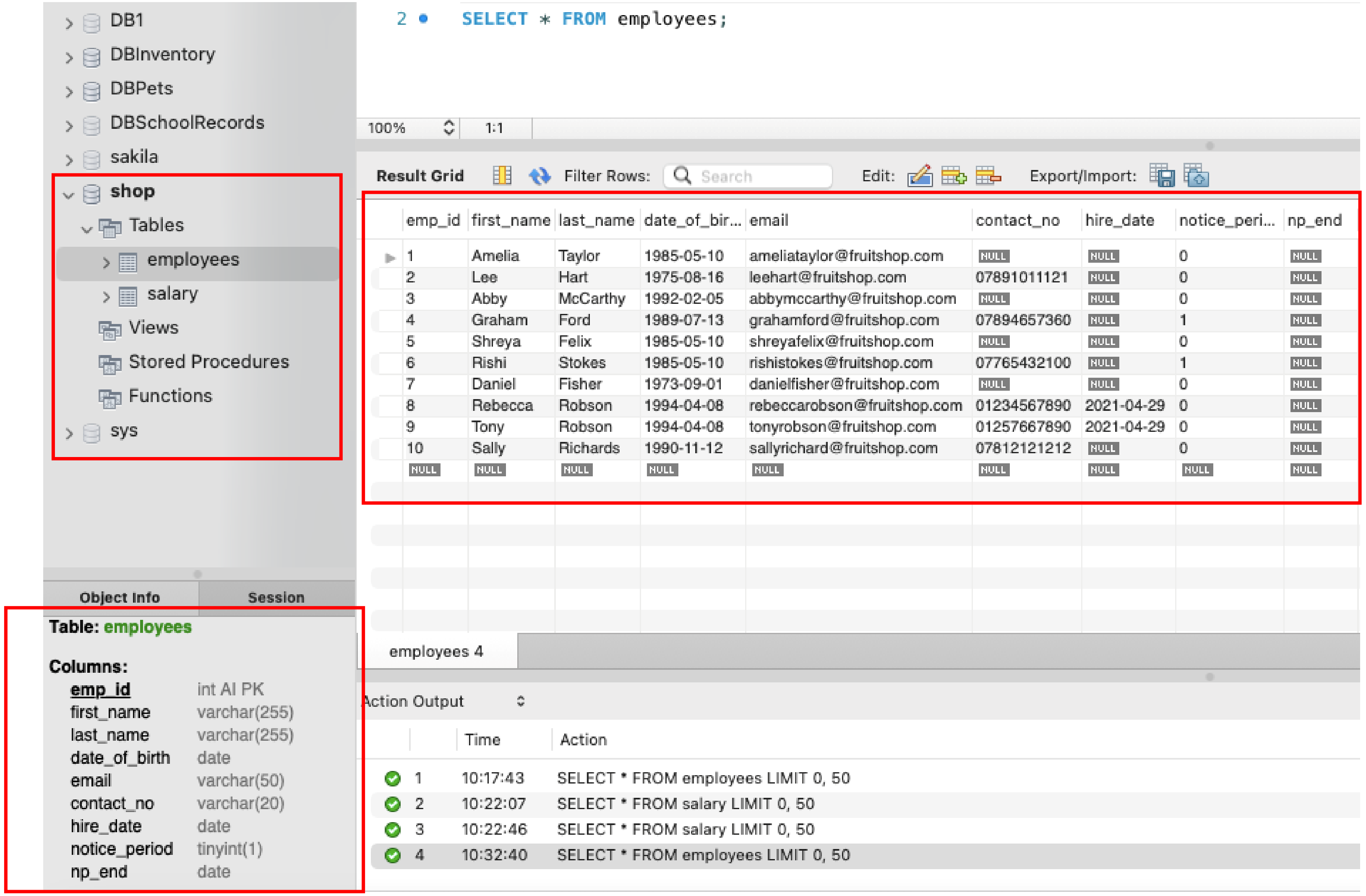
Table name: employees

```
4 •   SELECT * FROM salary;
5
```

emp_id	prior_salary	current_sala...
2	40000	44000
4	20000	25000
6	35000	40000
8	45000	45000
10	50000	53000
12	42000	43000
14	38000	42000
16	28000	35000
18	49000	56000
20	52000	58000

Table name: salary

- Check the data types of your columns before performing queries.



Steps of Joining Tables

- 1. Identify the tables you would like to join (you may provide aliases for simplicity).
- 2. Establish the type of join required.
- 3. Establish joining condition.
- 4. Choose columns of data you want in your single result.

Inner Join

Overview of the Database

```
2 ●   SELECT * FROM employees;
```

100% | 1:1

Result Grid | Filter Rows: Search | Edit: | Export/Import:

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_er
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Table name: employees

```
4 ●   SELECT * FROM salary;
5
```

100% | 22:2

Result Grid | Filter Rows:

emp_id	prior_salary	current_sala...
2	40000	44000
4	20000	25000
6	35000	40000
8	45000	45000
10	50000	53000
12	42000	43000
14	38000	42000
16	28000	35000
18	49000	56000
20	52000	58000

Table name: salary

MySQL Inner Join Syntax

- ```
  SELECT column_name(s)
  FROM table1
  INNER JOIN table2
  ON table1.column_name = table2.column_name;
  ```
  ```

Example 1

- Obtain employee ID, first name & last name, from the employee table and current salary from the salary table.

```sql
SELECT emp.emp_id,
           first_name,
           last_name,
           current_salary
FROM employees as emp
INNER JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

100% 19:2

Result Grid Filter Rows: Search

emp_id	first_name	last_name	current_sala...
2	Lee	Hart	44000
4	Graham	Ford	25000
6	Rishi	Stokes	40000
8	Rebecca	Robson	45000
10	Sally	Richards	53000

Example 2

- Obtain employee ID, first name & last name, from the employee table and prior and current salary from the salary table.

```sql
SELECT emp.emp_id,
            first_name,
            last_name,
            prior_salary,
            current_salary
FROM employees as emp
INNER JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

100% 24:2

Result Grid Filter Rows: Search

emp_id	first_name	last_name	prior_salary	current_sala...
2	Lee	Hart	40000	44000
4	Graham	Ford	20000	25000
6	Rishi	Stokes	35000	40000
8	Rebecca	Robson	45000	45000
10	Sally	Richards	50000	53000

Example 3

- Obtain first name & last name, from the employee table and prior and current salary from the salary table.

```
SELECT
        first_name,
        last_name,
        prior_salary,
        current_salary
FROM employees as emp
INNER JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

100% 22:3

Result Grid Filter Rows: Search

first_name	last_name	prior_salary	current_sala..
Lee	Hart	40000	44000
Graham	Ford	20000	25000
Rishi	Stokes	35000	40000
Rebecca	Robson	45000	45000
Sally	Richards	50000	53000

Example 4

- What if you did not specify the employee id in a specific column?

```sql
SELECT emp_id,
            first_name,
            last_name,
            prior_salary,
            current_salary
FROM employees as emp
INNER JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

		Time	Action	Response
	6	10:38:12	SELECT * FROM salary LIMIT 0, 50	10 row(s) returned
	7	10:55:42	SELECT emp.emp_id, first_name, last_name, current_salary FROM employees as emp INNER JOIN sal...	5 row(s) returned
	8	11:02:50	SELECT emp.emp_id, first_name, last_name, current_salary FROM employees as emp INNER JOIN sal...	5 row(s) returned
	9	11:03:05	SELECT emp.emp_id, first_name, last_name, current_salary FROM employees as emp INNER JOIN sal...	5 row(s) returned
	10	11:05:21	SELECT emp.emp_id, first_name, last_name, prior_salary, current_salary FROM employees as em...	5 row(s) returned
	11	11:06:30	SELECT first_name, last_name, prior_salary, current_salary FROM employees as emp INNER JOIN...	5 row(s) returned
	12	11:09:18	SELECT emp_id, first_name, last_name, prior_salary, current_salary FROM employees as emp INN...	Error Code: 1052. Column 'emp_id' in field list is ambi...

Error Code: 1052. Column 'emp_id' in field list is ambiguous

Left Join

Overview of the Database

```
2 •    SELECT * FROM employees;
```

100% 1:1

Result Grid Filter Rows: Search Edit: Export/Import:

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_en
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Table name: employees

```
4 •    SELECT * FROM salary;
5
```

100% 22:2

Result Grid Filter Rows:

emp_id	prior_salary	current_sala...
2	40000	44000
4	20000	25000
6	35000	40000
8	45000	45000
10	50000	53000
12	42000	43000
14	38000	42000
16	28000	35000
18	49000	56000
20	52000	58000

Table name: salary

MySQL Left Join Syntax

- ```
  SELECT column_name(s)
  FROM table1
  LEFT JOIN table2
  ON table1.column_name = table2.column_name;
  ```
  ```

Left Join

Example: Obtain all information from employees table and join salary table to bring back prior and current salary.

```sql
-- LEFT JOIN
SELECT emp.*,
       sal.prior_salary,
       sal.current_salary
FROM employees as emp
LEFT JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

100% 22:12

Result Grid Filter Rows: Search Export:

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_end	prior_salary	current_sala...
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL	40000	44000
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL	20000	25000
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL	35000	40000
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL	45000	45000
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL	NULL	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL	50000	53000

Result 15

Action Output

	Time	Action	Response	Duration / Fe
16	11:29:38	SELECT emp.*, sal.prior_salary, sal.current_salary FROM e...	10 row(s) returned	0.0011 sec / (

Key Notes

- 1. Use the LEFT JOIN operator syntax to join tables together – all information in the left table will feed the result.
- 2. Information from the right table will be populated where there is a math in the join condition
- 3. Where there is a no match, the right table values will be poplulated with null.

Right Join

MySQL Right Join Syntax

- ```
  SELECT column_name(s)
  FROM table1
  RIGHT JOIN table2
  ON table1.column_name = table2.column_name;
  ```
  ```

Overview of the Database

```
SELECT * FROM employees;
```

100% | 1:1

Result Grid | Filter Rows: Search | Edit: | Export/Import:

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_er
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Table name: employees

```
SELECT * FROM salary;
```

100% | 22:2

Result Grid | Filter Rows:

emp_id	prior_salary	current_sala...
2	40000	44000
4	20000	25000
6	35000	40000
8	45000	45000
10	50000	53000
12	42000	43000
14	38000	42000
16	28000	35000
18	49000	56000
20	52000	58000

Table name: salary

Example-Right Join

- Obtain all information from the salary table and join to the employees table to bring back the employees first name

```
-- RIGHT JOIN
SELECT  sal.*,
        emp.first_name
FROM employees as emp
RIGHT JOIN salary as sal
ON emp.first_name = sal.emp_id;
```

100% 17:19

Result Grid Filter Rows: Search Export:

emp_id	prior_salary	current_sala...	first_name
2	40000	44000	NULL
4	20000	25000	NULL
6	35000	40000	NULL
8	45000	45000	NULL
10	50000	53000	NULL
12	42000	43000	NULL
14	38000	42000	NULL
16	28000	35000	NULL
18	49000	56000	NULL
20	52000	58000	NULL

Result 16

Action Output

	Time	Action	Response
17	12:52:09	SELECT sal.*, emp.first_name FROM...	10 row(s) returned

Key Notes

- Right Table Join Example

```
SELECT *
FROM customer
RIGHT JOIN salary
ON customer.id = salary.customer_id;
```

The above query will return all columns from the left and right tables.
The rows from the right table will be present and populated with values from the left table where there is a match in the id from the customer table and the customer_id from the salary table.
Where there is no match, the left table columns will be populated with null.

Full Join

Overview of the Database

```sql
SELECT * FROM employees;
```

100% | 1:1

Result Grid | Filter Rows: Search | Edit: | Export/Import:

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_er
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Table name: employees

```sql
SELECT * FROM salary;
```

100% | 22:2

Result Grid | Filter Rows:

emp_id	prior_salary	current_sala...
2	40000	44000
4	20000	25000
6	35000	40000
8	45000	45000
10	50000	53000
12	42000	43000
14	38000	42000
16	28000	35000
18	49000	56000
20	52000	58000

Table name: salary

Full Outer Join Syntax

- MySQL does not directly support **FULL OUTER JOIN**, but you can simulate it using a combination of **LEFT JOIN** and **RIGHT JOIN** with a **UNION**, like this:

```
SELECT * FROM table1

LEFT JOIN table2 ON table1.id = table2.id

UNION

SELECT * FROM table1 RIGHT JOIN table2 ON
table1.id = table2.id;
```

Example-Full Outer Join

```sql
-- FULL OUTER JOIN-------
SELECT * FROM employees as emp
LEFT JOIN salary as sal
ON emp.emp_id = sal.emp_id
UNION
SELECT * FROM employees as emp
RIGHT JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

)0% 1:45

Result Grid Filter Rows: Search Export:

emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_end	emp_id	prior_salary	current_sala...
1	Amelia	Taylor	1985-05-10	ameliataylor@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL	NULL
2	Lee	Hart	1975-08-16	leehart@fruitshop.com	07891011121	NULL	0	NULL	2	40000	44000
3	Abby	McCarthy	1992-02-05	abbymccarthy@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL	NULL
4	Graham	Ford	1989-07-13	grahamford@fruitshop.com	07894657360	NULL	1	NULL	4	20000	25000
5	Shreya	Felix	1985-05-10	shreyafelix@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL	NULL
6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL	6	35000	40000
7	Daniel	Fisher	1973-09-01	danielfisher@fruitshop.com	NULL	NULL	0	NULL	NULL	NULL	NULL
8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL	8	45000	45000
9	Tony	Robson	1994-04-08	tonyrobson@fruitshop.com	01257667890	2021-04-29	0	NULL	NULL	NULL	NULL
10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL	10	50000	53000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	12	42000	43000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	14	38000	42000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	16	28000	35000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	18	49000	56000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	20	52000	58000

Result 27

tion Output

	Time	Action	Response
31	13:35:00	SELECT * FROM employees as emp LEFT JOIN salary as sal ON emp.emp_id = sal.emp_id UNION SELE...	15 row(s) returned

Example Full Outer Join with Redundancy

```sql
-- FULL OUTER JOIN-----------
SELECT * FROM salary as sal
LEFT JOIN employees as emp
ON emp.emp_id = sal.emp_id
UNION
SELECT * FROM employees as emp
RIGHT JOIN salary as sal
ON emp.emp_id = sal.emp_id;
```

100% 1:55

Result Grid Filter Rows: Search Export:

emp_id	prior_salary	current_sala...	emp_id	first_name	last_name	date_of_bir...	email	contact_no	hire_date	notice_peri...	np_end
6	35000	40000	6	Rishi	Stokes	1985-05-10	rishistokes@fruitshop.com	07765432100	NULL	1	NULL
8	45000	45000	8	Rebecca	Robson	1994-04-08	rebeccarobson@fruitshop.com	01234567890	2021-04-29	0	NULL
10	50000	53000	10	Sally	Richards	1990-11-12	sallyrichard@fruitshop.com	07812121212	NULL	0	NULL
12	42000	43000	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
14	38000	42000	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
16	28000	35000	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
18	49000	56000	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
20	52000	58000	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
2	Lee	Hart	1975-...	leehart@...	0789101...	NULL	0	NULL	2	40000	44000
4	Graham	Ford	1989-...	grahamf...	0789465...	NULL	1	NULL	4	20000	25000
6	Rishi	Stokes	1985-...	rishistok...	0776543...	NULL	1	NULL	6	35000	40000
8	Rebecca	Robson	1994-...	rebeccar...	0123456...	2021-04-29	0	NULL	8	45000	45000
10	Sally	Richards	1990-...	sallyricha...	0781212...	NULL	0	NULL	10	50000	53000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	12	42000	43000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	14	38000	42000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	16	28000	35000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	18	49000	56000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	20	52000	58000

Result 29

Action Output

	Time	Action	Response
33	13:37:30	SELECT * FROM salary as sal LEFT JOIN employees as emp ON emp.emp_id = sal.emp_id UNION SELE...	20 row(s) returned